CS52 · AI & ML · CIE-1 · PREDICTED QUESTIONS

Unit II: Logic Solving, Bayes & Intro to ML

Curiosity killed the cat · ML case-based (S / U / R) · Bayes reasoning · John is happy · John likes peanuts · Task / Performance / Experience · Find-S

SURE SHOT named in the insider list / CR list or repeated from last year — learn word-perfect

VERY LIKELY came last year or is the natural variant

IMPORTANT TOPIC fits the "scenario" hint — study the idea

II Unit II — Logic Solving, Bayes & Intro to ML

Q1. "Curiosity killed the cat" — prove using resolution. SURE SHOT

Why: named directly on the insider list. It's the classic textbook resolution problem — learn it fully (it can carry 7–8 marks).

QUESTION

Jack owns a dog. Every dog owner is an animal lover. No animal lover kills an animal. Either Jack or Curiosity killed the cat, who is named Tuna. Did Curiosity kill the cat?

Step 1 — English → FOL

A. ∃x Dog(x) ∧ Owns(Jack, x)

B. ∀x (∃y Dog(y) ∧ Owns(x, y)) ⇒ AnimalLover(x)

C. ∀x AnimalLover(x) ⇒ (∀y Animal(y) ⇒ ¬Kills(x, y))

D. Kills(Jack, Tuna) ∨ Kills(Curiosity, Tuna)

E. Cat(Tuna)

F. ∀x Cat(x) ⇒ Animal(x) (background: cats are animals)

Goal: Kills(Curiosity, Tuna)

Step 2 — CNF (clause form)

A1. Dog(D) A2. Owns(Jack, D) (EI: Skolem constant D)

B. ¬Dog(y) ∨ ¬Owns(x, y) ∨ AnimalLover(x)

C. ¬AnimalLover(a) ∨ ¬Animal(b) ∨ ¬Kills(a, b)

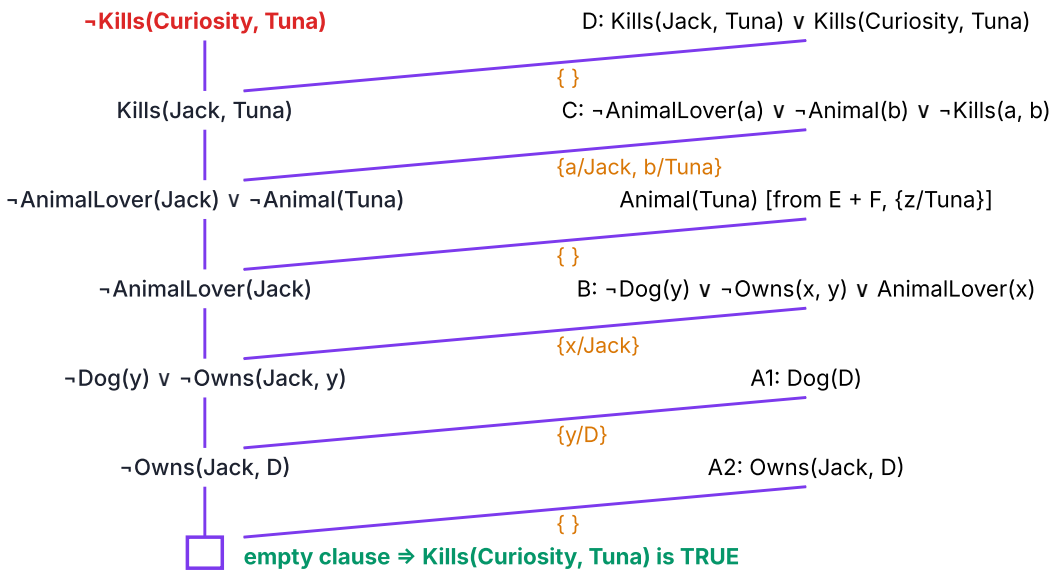
D. Kills(Jack, Tuna) ∨ Kills(Curiosity, Tuna)

E. Cat(Tuna)

F. ¬Cat(z) ∨ Animal(z)

Step 3 — negate goal: ¬Kills(Curiosity, Tuna)

How B became a clause: (∃y Dog(y) ∧ Owns(x, y)) ⇒ AnimalLover(x) → ¬∃y(...) ∨ ... → ∀y (¬Dog(y) ∨ ¬Owns(x, y)) ∨ AnimalLover(x). **C:** ¬AnimalLover(x) ∨ ∀y(¬Animal(y) ∨ ¬Kills(x, y)), variables renamed a, b (standardize apart).



Step 4 — resolution graph (left: resolvents; right: clause used; orange: substitution). "Animal(Tuna)" first comes from resolving E with F using {z/Tuna}.

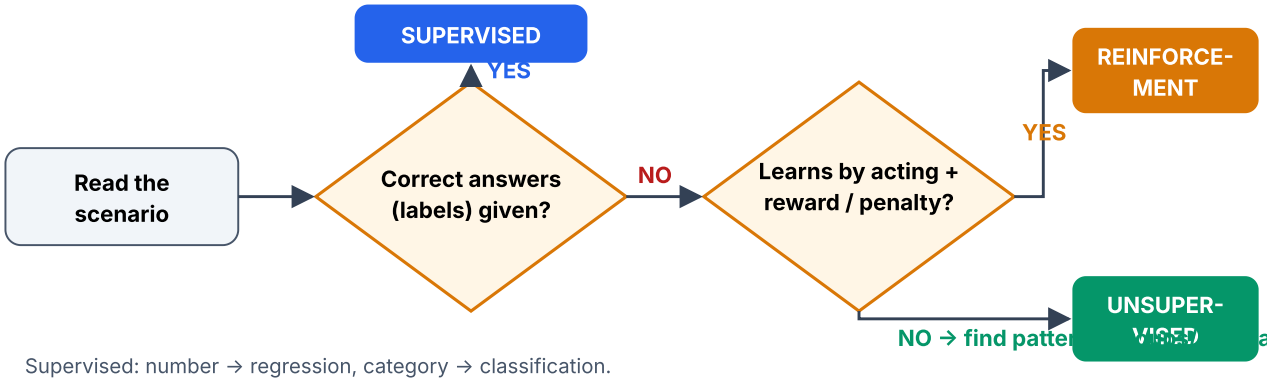
Conclusion: ¬Kills(Curiosity, Tuna) leads to a contradiction, so **Curiosity killed the cat**. (Reason in English: Jack owns a dog → Jack is an animal lover → he can't kill an animal → Tuna is a cat, so an animal → Jack didn't kill Tuna → by D, Curiosity did.)

REMEMBER IT — THE STORY CHAIN

Say it as one breath: "**Dog → Lover → No-kill → Cat is Animal → Not Jack → Curiosity**." Resolution just walks this chain backwards from ¬Kills(Curiosity, Tuna). Six resolve steps; the two Skolem facts Dog(D), Owns(Jack, D) are used last.

Q2. Identify the type of learning (supervised / unsupervised / reinforcement) in each scenario and justify. SURE SHOT

Why: last year's Q1c (5 marks) + "Machine Learning → case based example" on the insider list. Expect 5 new scenarios — use the flowchart.



Decision flowchart — ask two questions and you're done.

Scenario (last year's paper)	Type	One-line justification
A. Hospital uses patient records with labelled diagnoses to predict diseases in new patients	**Supervised (classification)**	Labelled examples (record → diagnosis) are given
B. Game AI maximises chess score by trying strategies, learning from wins/losses/draws	**Reinforcement**	Learns by trial and error from reward (win) / penalty (loss)
C. Company analyses sales data to detect anomalies (sudden drops/spikes) without knowing causes	**Unsupervised** (anomaly detection)	No labels; finds unusual patterns
D. Self-driving car learns decisions by interacting with a simulator and receiving rewards/penalties	**Reinforcement**	Agent–environment interaction + rewards
E. Customer reviews fed to an algorithm that finds distinct topics; no labels	**Unsupervised** (clustering / topic modelling)	Groups similar reviews without labels

Practice set — likely new scenarios this year		
Spam filter trained on emails marked spam / not spam	Supervised – classification	labels given
Predict house prices from past sales	Supervised – regression	numeric label
Group customers into segments for marketing	Unsupervised – clustering	no labels
"Customers who bought bread also bought butter"	Unsupervised – association	finds co-occurrence
Robot learns to walk by falling and getting penalties	Reinforcement	reward/penalty
Bank flags fraud using past transactions labelled fraud/genuine	Supervised – classification	labels given
Detect unusual network traffic with no attack labels	Unsupervised – anomaly detection	no labels
Thermostat learns to save energy from comfort feedback	Reinforcement	feedback signal, not correct answers
Face unlock trained on labelled photos of the owner	Supervised	labelled images
Compress 100 features into 3 for visualisation	Unsupervised – dimensionality reduction (PCA)	no labels

ANSWER FORMAT (1 MARK EACH)

Write "**A → Supervised.** Because labelled data (diagnosis) is available; the model learns input → output mapping (classification)." Always include the **keyword from the scenario** that decided it ("labelled", "reward", "no prior labels").

REMEMBER IT

Teacher, Explorer, Gamer. Supervised = learns from a **teacher** who gives answers. Unsupervised = an **explorer** who finds groups alone. Reinforcement = a **gamer** who learns from score / reward. Keywords: "labelled" → S; "no labels / patterns / groups / anomalies" → U; "reward / penalty / trial-and-error / interacts" → R.

Q3. Bayes' theorem reasoning: a test is positive — what's the probability the person really has the disease? SURE SHOT

Why: "Bayesian theorem reasoning" on the insider list. Expect a medical / spam / fraud scenario with numbers. Same 4-line method every time.

TYPICAL QUESTION

A disease affects 1% of people. A test detects it 95% of the time when present (sensitivity), but also gives a false positive for 10% of healthy people. A patient tests positive. Find P(Disease | Positive). Should the doctor conclude the patient has the disease?

Given: $P(D) = 0.01$, $P(\neg D) = 0.99$
$P(+|D) = 0.95$, $P(+|\neg D) = 0.10$
Bayes: $P(D|+) = P(+|D)\,P(D) / P(+)$
Total probability: $P(+) = P(+|D)P(D) + P(+|\neg D)P(\neg D)$
$= 0.95 \times 0.01 + 0.10 \times 0.99 = 0.0095 + 0.099 = 0.1085$
$P(D|+) = 0.0095 / 0.1085 =$ **0.0876 ≈ 8.8%**

Reasoning (the "why" mark): even with a positive test there's only ~9% chance of disease, because the disease is **rare** (prior 1%) and false positives from the 99% healthy people outnumber true positives (0.099 vs 0.0095). A strong test doesn't mean a high posterior — the prior matters. The doctor should order a second, confirmatory test.

MAP form (if asked "diagnose"): compare $P(+|D)P(D) = 0.0095$ vs $P(+|\neg D)P(\neg D) = 0.099$ → h_{MAP} = ¬Disease.

Same method, other classic numbers (know the answers)	
Cancer test: P(cancer) = 0.008, P(+|cancer) = 0.98, P(+|¬cancer) = 0.03	0.0078 vs 0.0298 → ¬cancer; P(cancer|+) = 0.21
Meningitis: P(S|M) = 0.7, P(M) = 1/50000, P(S) = 0.01	P(M|S) = 0.0014
Spam: P(S) = 0.2, P(offer|S) = 0.8, P(offer) = 0.4	P(S|offer) = 0.40
Liver disease: P(A) = 0.10, P(B) = 0.05, P(B|A) = 0.07	P(A|B) = 0.14
Face card: P(King | Face)	(1 × 4/52) / (12/52) = 1/3

REMEMBER IT

"Post = Like × Prior ÷ Evidence" ("**P**osting **L**ikes **P**leases **E**veryone"). Steps: **G-B-T-A** — write **G**ivens + complements → **B**ayes formula → **T**otal probability for the denominator → **A**nswer + one-line meaning.

Q4. Using resolution refutation, check whether "John is happy". VERY LIKELY

Why: last year's Q3b (8 marks). If Curiosity comes as the main resolution question, this may come as the alternate choice.

QUESTION

Anyone passing his history exams and winning the lottery is happy. But anyone who studies or is lucky can pass all his exams. John did not study, but John is lucky. Anyone who is lucky wins the lottery.

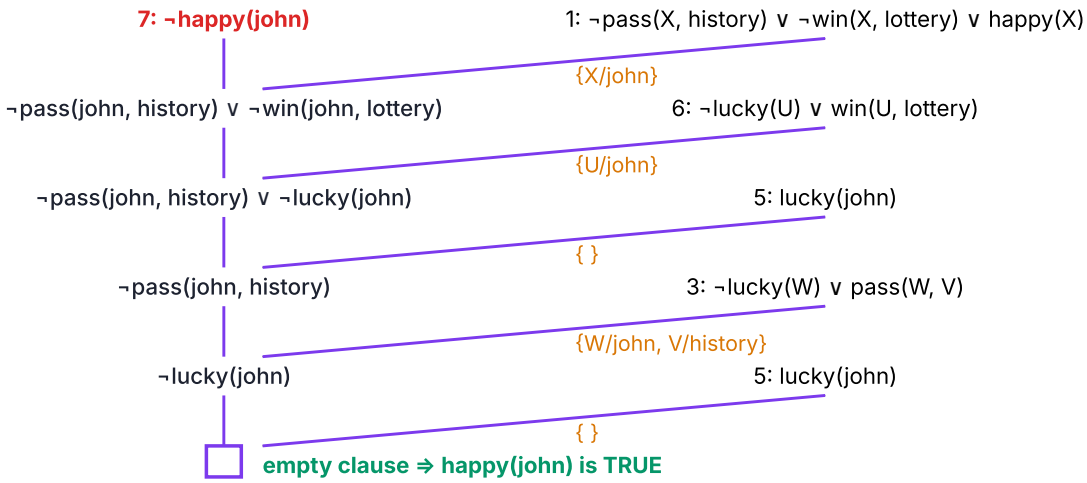
FOL

$\forall X\ pass(X, history) \wedge win(X, lottery) \Rightarrow happy(X)$
$\forall X\ \forall Y\ study(X) \vee lucky(X) \Rightarrow pass(X, Y)$
$\neg study(john) \wedge lucky(john)$
$\forall X\ lucky(X) \Rightarrow win(X, lottery)$

Clauses

1. $\neg pass(X, history) \vee \neg win(X, lottery) \vee happy(X)$
2. $\neg study(Y) \vee pass(Y, Z)$ 3. $\neg lucky(W) \vee pass(W, V)$
4. $\neg study(john)$ 5. $lucky(john)$
6. $\neg lucky(U) \vee win(U, lottery)$
7. **$\neg happy(john)$** (negated goal)

Clauses 2 & 3 come from ¬(study ∨ lucky) ∨ pass → (¬study ∨ pass) ∧ (¬lucky ∨ pass).



REMEMBER IT

"Happy needs Pass + Win; Lucky gives both." So the proof uses lucky(john) **twice** — once to win the lottery, once to pass history. Clause 2 and 4 (study) are a distraction — never used.

Q5. Translate into predicate logic and prove "John likes peanuts" using backward (or forward) chaining. VERY LIKELY

Why: last year's Q2b (7 marks). A repeat or the forward-chaining version is common.

(i) Translation

1. John likes all kinds of food: ∀x Food(x) ⇒ Likes(John, x)
2. Apples are food: Food(Apple)
3. Chicken is food: Food(Chicken)
4. Anything anyone eats and isn't killed by is food: ∀x ∀y Eats(x, y) ∧ ¬Killed(x) ⇒ Food(y)
5. Bill eats peanuts and is still alive: Eats(Bill, Peanut) ∧ ¬Killed(Bill)
6. Sue eats everything Bill eats: ∀x Eats(Bill, x) ⇒ Eats(Sue, x)

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(ii) Backward chaining — goal first
Goal: Likes(John, Peanut)
 └ rule 1 {x/Peanut} → Food(Peanut)
    └ rule 4 {y/Peanut} → Eats(x, Peanut) ∧ ¬Killed(x)
       ├ Eats(x, Peanut) {x/Bill} → fact 5 ✔
       └ ¬Killed(Bill)            → fact 5 ✔
 ⇒ Food(Peanut) ✔ ⇒ Likes(John, Peanut) ✔

Forward chaining (if asked):
 facts → rule 6: Eats(Sue, Peanut)
       → rule 4: Food(Peanut)
       → rule 1: Likes(John, Peanut) ✔
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EXAM TIPS

Draw the backward-chaining tree **top-down** (goal at top, facts at leaves, ✔ on each fact, substitution on each edge). Forward chaining is **bottom-up** (facts first). One line to define each: BC = goal-driven, FC = data-driven.

REMEMBER IT

"Likes ← Food ← Eats + Alive." Bill is the only one with both facts, so x = Bill. Apples, Chicken and Sue are distractions.

Q6. Identify Task (T), Performance (P) and Experience (E) for a given learning problem (well-posed learning problem). IMPORTANT TOPIC

Why: scenario-based, fits sir's hint and CO3 ("fundamental understanding of ML"). Quick 5 marks.

DEFINITION (TOM MITCHELL)

A program learns from **experience E** with respect to a class of **tasks T** and **performance measure P** if its performance at T, as measured by P, improves with E. A problem is well-posed when T, P and E are all clearly identified.

Problem	T	P	E
Checkers	Playing checkers	% games won	Games played against itself
Handwriting recognition	Recognise handwritten words in images	% words correctly classified	Database of labelled handwritten words
Robot driving	Drive on 4-lane highways using vision	Average distance before an error	Images + steering commands recorded from a human driver
Spam filter	Classify emails as spam / not	% correctly classified	Emails labelled by users
Movie recommender	Recommend movies a user will like	% recommendations watched / rated high	User's past ratings and watch history
Crop-disease detector	Identify disease from leaf photos	% correct diagnoses	Labelled leaf images from farmers

REMEMBER IT

T = verb (what it does), **P = percentage/number** (how we score it), **E = data** (what it learns from). If your P isn't a number, it's wrong.

Q7. Apply Find-S to find the maximally specific hypothesis (concept learning).

IMPORTANT TOPIC

Why: CO3 last year explicitly said "Concept learning". It's a table problem (not pseudocode), so it fits "problem solving".

Sky	AirTemp	Humidity	Wind	Water	Forecast	EnjoySport
Sunny	Warm	Normal	Strong	Warm	Same	**Yes**
Sunny	Warm	High	Strong	Warm	Same	**Yes**
Rainy	Cold	High	Strong	Warm	Change	**No**
Sunny	Warm	High	Strong	Cool	Change	**Yes**

h0 = ⟨Ø, Ø, Ø, Ø, Ø, Ø⟩ (most specific)

x1 (+): h1 = ⟨Sunny, Warm, Normal, Strong, Warm, Same⟩ — copy it

x2 (+): Humidity differs → ? h2 = ⟨Sunny, Warm, ?, Strong, Warm, Same⟩

x3 (−): ignored (Find-S skips negatives)

x4 (+): Water, Forecast differ → h4 = **⟨Sunny, Warm, ?, Strong, ?, ?⟩**

Limitations (1 mark): ignores negative examples; can't detect noisy data; gives only one hypothesis (Candidate-Elimination gives the whole version space: S = {⟨Sunny, Warm, ?, Strong, ?, ?⟩}, G = {⟨Sunny, ?, ?, ?, ?, ?⟩, ⟨?, Warm, ?, ?, ?, ?⟩}).

REMEMBER IT

"Copy the first Yes, then Question-mark the differences, Skip the No's." (C-Q-S)

UNIT II — ALSO BE READY FOR (STUDY THE TOPIC)

English → FOL (∀ with ⇒, ∃ with ∧) · unification / MGU of two expressions · forward chaining "West is a criminal" · Bayesian network joint probability P = ∏ P(node | parents) · knowledge graph RDF triple (subject–predicate–object) and KG vs ontology.